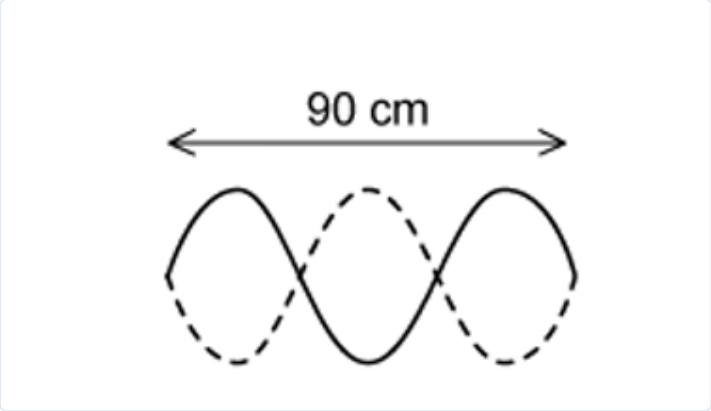
CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

8.1 Stationary Waves

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 31 Questions (31 Marks)

Question 1 Easy [1 mark]

A stationary wave shown in the diagram was created on a stretched string, it had two instants of maximum vertical displacement.



The frequency of the wave is 13 Hz.

What is the speed of the wave?

A 3.9 m s^{-1}

B 7.8 m s^{-1}

C 390 m s^{-1}

D 780 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Wave speed = frequency × wavelength
  - $v = f \quad \lambda$
- Frequency = 13 Hz
- Wavelength: in the diagram there are 1.5 waves in 90 cm
  - $\frac{90}{1.5} = 60 \quad \text{cm} = 0.6 \quad \text{m}$
- Therefore, $v = 13 \times 0.6 = 7.8 \quad \text{m} \quad \text{s}^{-1}$

Question 2 Easy [1 mark]

The table shows some statements about stationary and progressive waves.

Which row is correct ?

	Progressive wave	Stationary wave
A	particles one wavelength apart vibrate in phase	particles one wavelength apart vibrate in phase
B	particles vibrate in phase with their immediate neighbours	particles in adjacent loops vibrate in antiphase
C	energy is transferred along the wave	energy is transferred along the wave
D	all particles vibrate with the same amplitude	all particles vibrate with the same amplitude

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- In a progressive wave, the wavefronts move through the medium transferring energy from one point to the next
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
    - The waves in a standing wave are one wavelength apart and vibrate in phase
    - Particles in a progressive wave can also be one wavelength apart and vibrate in phase

B is incorrect because in a standing wave, adjacent particles are vibrating in phase with each other

C is incorrect because only a progressive wave transfers energy, in a standing wave there is no net transfer of energy

D is incorrect because in a standing wave adjacent points each oscillate with a different amplitude

Question 3 Easy [1 mark]

When a stationary wave is produced in a tube, which of the following statements describes the wave speed?

A it is the speed of a particle at a node

B it is the speed of one of the progressive waves that are producing the stationary wave

C it is the speed at which energy is transferred from one antinode to an adjacent antinode

D it is the distance between two adjacent nodes divided by the period of the wave

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In a progressive wave, the wavefronts move through the medium transferring energy from one point to the next
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The wave speed of a standing wave is the speed that one of the progressive wave travels that is producing the standing wave

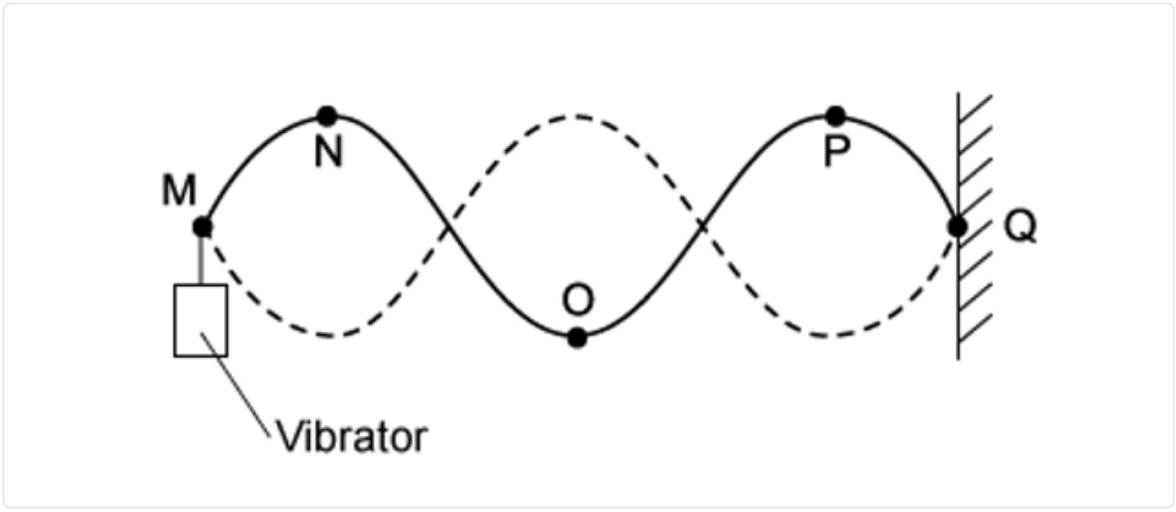
A is incorrect because the wave speed is, $v = f \quad \lambda$, at a node the wave speed = 0

C is incorrect because a standing wave does not transfer energy

D is incorrect because the distance between two antinodes would be the wavelength of the wave. The period of the wave is the time it takes for one wave to travel

Question 4 Easy [1 mark]

A stretched string is set up to demonstrate a stationary wave between two points M and Q.



Which of the following statements are correct?

A the distance between M and Q is three wavelengths

B the wave shown has the lowest possible frequency

C points N and P vibrate in phase

D point O is at a node

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- N and P are vibrating in phase
    - This means they are vibrating up and down at the same time

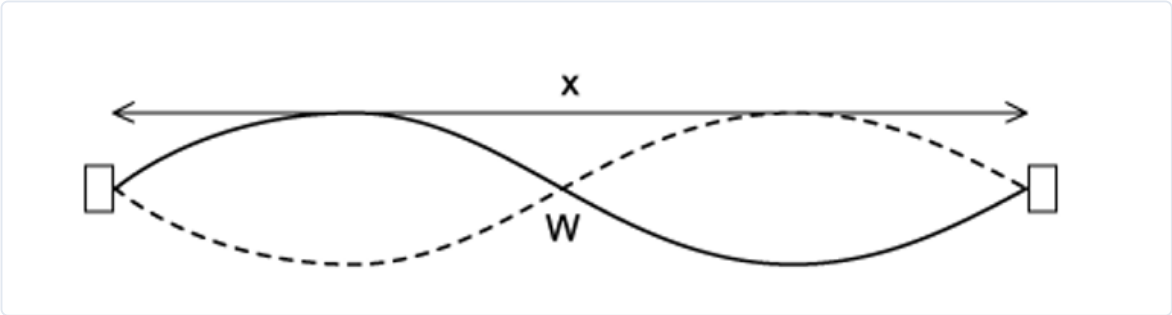
A is incorrect because there are 1.5 wavelengths between M and Q

B is incorrect because a wave with the lowest frequency will be half a wave

D is incorrect because point O is at an antinode

Question 5 Easy [1 mark]

A stretched string is used to demonstrate a stationary wave, as shown in the diagram.



Which row in the table correctly describes the length of x and the name of W?

	Length x	Point W
A	two wavelengths	node
B	one wavelength	node
C	two wavelengths	antinode
D	one wavelength	antinode

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

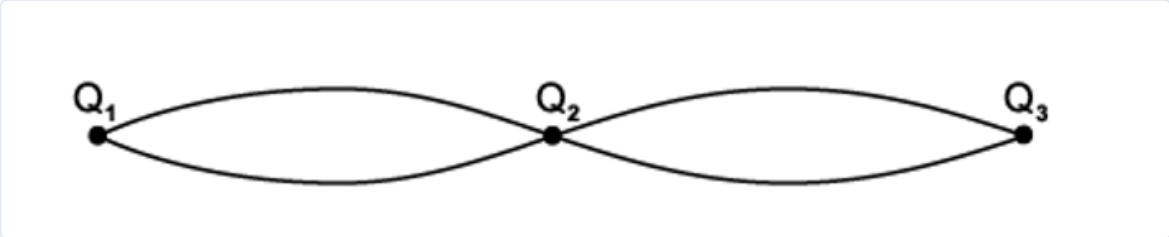
- A wavelength is the length of one complete wave
  - Each loop is the equivalent of one-half of a wavelength
- A node is the point on a standing wave of minimum amplitude

A & **C** are incorrect because there are not two wavelengths in the diagram

D is incorrect because the antinode is the point of maximum amplitude of the wave and found at the crest of the loop

Question 6 Easy [1 mark]

A stationary wave on a string is shown in the diagram below. The standing wave has three nodes Q_1, Q_2 and Q_3



Which of the following statements are correct ?

A all points on the string vibrate in phase

B all points on the string vibrate with the same amplitude

C points equidistant from Q_2 vibrate with the same frequency and the same amplitude

D points equidistant from Q_2 vibrate with the same frequency and in phase

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The points that are equidistant from Q_2 will vibrate with the same frequency and amplitude

A is incorrect because not all the points on the wave vibrate in phase. Points within the loops will vibrate in phase but vibrate out of phase with the adjacent loop

B is incorrect because not all the points on the string vibrate with the same amplitude. Points closer to the nodes have a smaller amplitude than those as the crest of the loop

D is incorrect because points that an equidistant from Q_2 are not in phase as points in adjacent loops are out of phase

Question 7 Easy [1 mark]

A student formed a stationary wave in a tube by blowing over the top as shown in the diagram



Which of the following statements are correct ?

A the stationary wave in the cylinder is polarised

B the stationary wave will have an antinode at the bottom of the cylinder

C the fundamental frequency of the stationary wave decreases when some water is added to the cylinder

D the stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A standing wave is one in which two waves combine constructively giving an amplitude of twice the original wave
    - Therefore, the stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions

A is incorrect because sound waves can't be polarised

B is incorrect because the waves will meet at a node at the fixed end of the cylinder

C is incorrect because decreasing the length of the column (done by adding water to the cylinder) will increase the fundamental frequency of the system

Question 8 Easy [1 mark]

A stationary wave is formed when two waves travel in opposite directions through the same space.

Which of the following statements must also be correct?

A the waves must have equal speed, frequency and wavelength

B the waves must have equal frequency, but a different speed and wavelength

C the waves must have equal wavelength, but a different speed and frequency

D the waves must have equal speed, but a different wavelength and frequency

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A standing wave is one in which two waves combine constructively giving an amplitude of twice the original wave
    - Therefore, the waves must have equal speed, frequency and wavelength

B & **D** are incorrect because if the waves had a different wavelengths, then they would not be in phase

C is incorrect because the waves must have the same frequency to create a standing wave

Question 9 Easy [1 mark]

A student sets up a sound wave in a long tube, and the tube is closed at one end. The tube length is changed until the sound from the tube is loudest.

Which of the following statements describe the nature of the wave in the tube?

A transverse and stationary

B transverse and progressive

C longitudinal and stationary

D longitudinal and progressive

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The wave is a sound wave, which is a longitudinal wave
    - Longitudinal waves travel parallel to the direction of travel
- The wave created by constructive combining of the sound waves created the largest amplitude, so the loudest sound
    - Hence the wave created is a stationary wave

A & **B** are incorrect because sound waves are not transverse waves; transverse waves travel perpendicular to the direction of travel

D is incorrect because superposition of the wave has occurred creating a standing wave, so the wave is not progressive

Question 10 Easy [1 mark]

A student wants to demonstrate a stationary wave.

What could they use to produce this?

A passing water waves through a narrow slit

B passing monochromatic light through a double slit

C making a loud sound near a mountain

D blowing air over the top of an empty bottle

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- When air is blown over the top of an empty bottle, the bottom of the bottle has a fixed end, creating a wave travelling in the opposite direction with the same frequency and wavelength

A is incorrect because a single slit would diffract the wave into semicircular waves

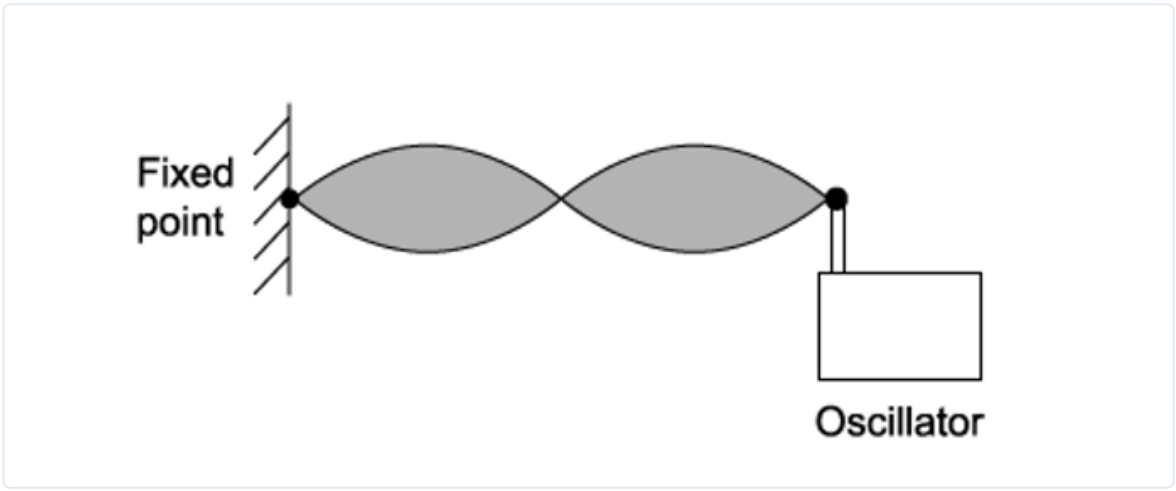
B is incorrect because this would create the characteristic fringes caused by the superposition of the overlapping light from the two slits

C is incorrect because this would create an echo, which is a reflection of a sound wave

Question 11 Medium [1 mark]

A student was investigating the speed of a transverse wave on a stretched spring. They found that by adjusting the tension on the spring, they were able to change the speed.

The wave pattern produced is shown in the diagram the frequency of the oscillator was set to 650 Hz



What needs to be changed to maintain the same wave pattern when the frequency is increased to 750 Hz?

A increase the speed of the wave on the string

B increase the wavelength of the wave on the string

C decrease the wavelength of the wave on the string

D decrease the speed of the wave on the string

Mark Scheme & Explanation:

*The correct answer is **A** because:*

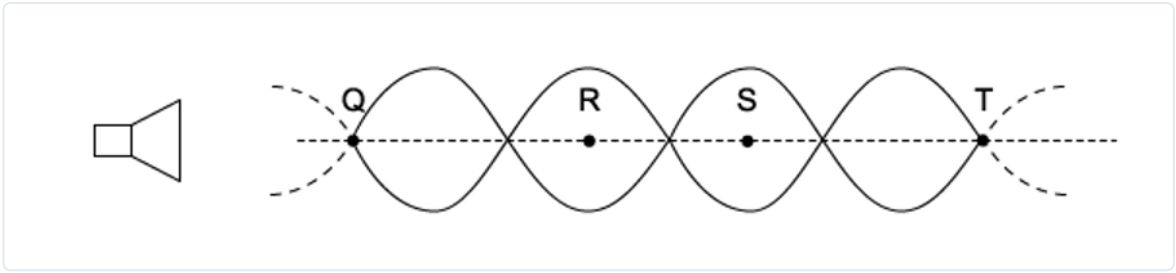
- The diagram shows one complete wave is produced with a frequency of 650 Hz
  - The wave equation is given by $v = f\lambda$
- In order for the wavelength to stay the same as the frequency increases, the speed of the wave also needs to be increased

B & **C** are incorrect because in order to maintain the same wave pattern, the wavelength must stay the same

D is incorrect because if the speed is decreased, then the wavelength will decrease changing the wave pattern

Question 12 Medium [1 mark]

The pattern produced when two waves are produced by superposition of sound waves from a loudspeaker reflected by a metal sheet.



Q, R, S and T are four points on the line through the centre of these waves.

Which of the following statements is correct?

A a node is a quarter of a wavelength from an adjacent antinode

B the stationary waves oscillate at right angles to the line QT

C an antinode is formed at the surface of the metal sheet

D the oscillations at R are in phase with those at S

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A node is produced with zero displacement and an antinode at maximum displacement
- The distance between two nodes is half a wavelength, and there is an antinode at the middle of two successive nodes
    - Therefore, a node is a quarter of a wavelength from an adjacent antinode

B is incorrect because stationary waves do not oscillate

C is incorrect because a node would be formed at the metal sheet

D is incorrect because in a stationary wave, the points a loop are in phase and out of phase with adjacent loops

Question 13 Medium [1 mark]

Headphones that have noise reduction can cancel out external noise by producing their own waves.

They are fitted with a microphone that detects the external sound frequency. The loudspeaker then produces a wave with the same frequency but different phase.

What would the phase difference needed to cancel out the external sound?

A 360°

B 270°

C 180°

D 90°

Mark Scheme & Explanation:

*The correct answer is **C** because:*

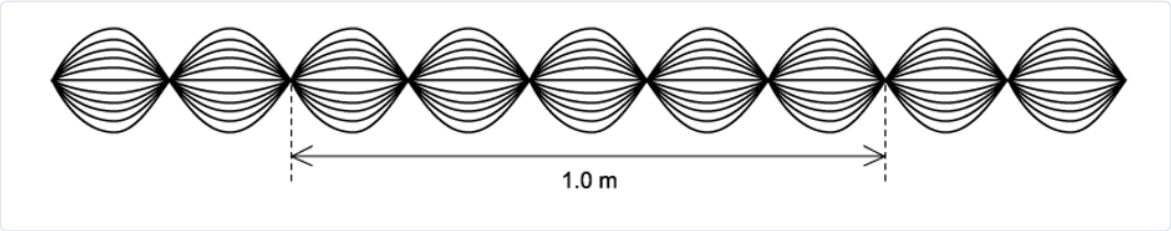
- The principle of superposition is applied when two or more waves are travelling through the same medium at the same time
- Noise-cancelling headphones use destructive interference to cancel out the external noise
- Therefore, the headphone will need to produce a wave with the same frequency and amplitude but with a phase difference of 180°

A is incorrect because a phase difference of 360° will mean that the waves are in phase and this will cause constructive interference making the external noise louder

B & **D** are incorrect because a phase difference of 90° and 270° will cause a comb filter effect, alternating constructive and destructive peaks and troughs

Question 14 Medium [1 mark]

The diagram below shows a wave pattern over time



What is the correct description of the wave?

A the wave is transverse, has a wavelength of 40 cm and is stationary

B the wave is transverse, has a wavelength of 40 cm and is progressive

C the wave is transverse, has a wavelength of 20 cm and is stationary

D the wave is longitudinal, has a wavelength of 20 cm and is stationary

Mark Scheme & Explanation:

*The correct answer is **A** because:*

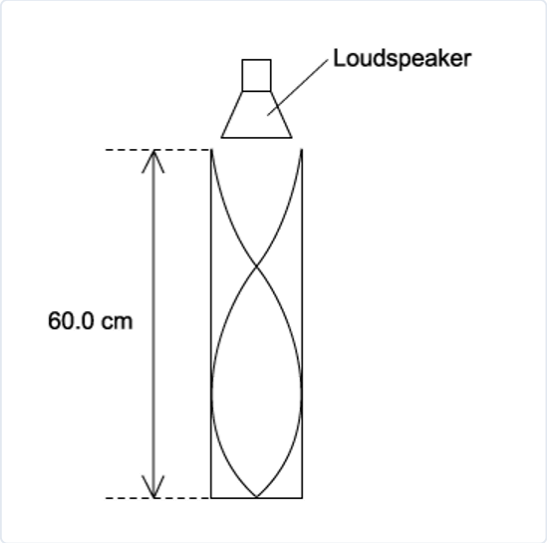
- The diagram is a transverse wave as the displacement is vertical and the direction of travel is horizontal
- 1 loop is half a wave
- 100 cm has 2.5 waves
  - $100 = 2.5\lambda$
  - $\lambda = \frac{100}{2.5} = 40 \text{ cm}$
- So the wavelength is 40 cm
- There are nodes present that remain still, hence it is a stationary wave

B is incorrect because the diagram shows a stationary wave, not a progressive wave

C & **D** are incorrect because there are 2.5 waves in 100 cm, so the wavelength is 40 cm, not 20 cm

Question 15 Medium [1 mark]

The diagram below shows sound wave produced in a column by a loudspeaker. The speed of sound in air is 330 m s^{-1}



What is the frequency of the wave?

A 1650 Hz

B 830 Hz

C 550 Hz

D 413 Hz

Mark Scheme & Explanation:

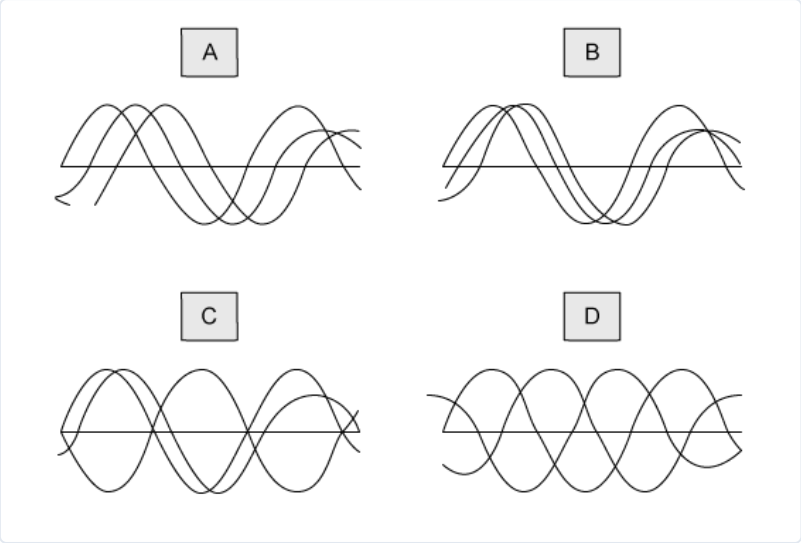
*The correct answer is **D** because:*

- The diagram shows $\frac{3}{4}$ of a wavelength is formed in a 60 cm tube
  - Therefore, one wavelength is equal to $\lambda = 80 \text{ cm} = 0.8 \text{ m}$
- The wave equation is given by $v = f \lambda$
  - $f = \frac{v}{\lambda} = \frac{330}{0.8} = 412.5 \text{ Hz}$
- Therefore, the frequency of the wave is 413 Hz

Question 16 Medium [1 mark]

Each of the diagrams below show three waves; each of the waves have the same amplitude and frequency but have different phases.

The waves are added together to give a resultant wave. In which case would the resultant wave be zero ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- In order for the displacement to be zero the sum of the displacements needs to be zero
- One wave has zero displacement, one negative and one positive, giving a resultant displacement of zero

A & **B** are incorrect because they both give a negative displacement

C is incorrect because two of the three waves will have a negative displacement but will still leave one wave; therefore the resultant wave will not be zero

Question 17 Medium [1 mark]

A stationary sound wave produced in a tube has a series of nodes. The distance between the first node and the sixth node is 30 cm

What is the wavelength of the wave ?

A 12.0 cm

B 10.0 cm

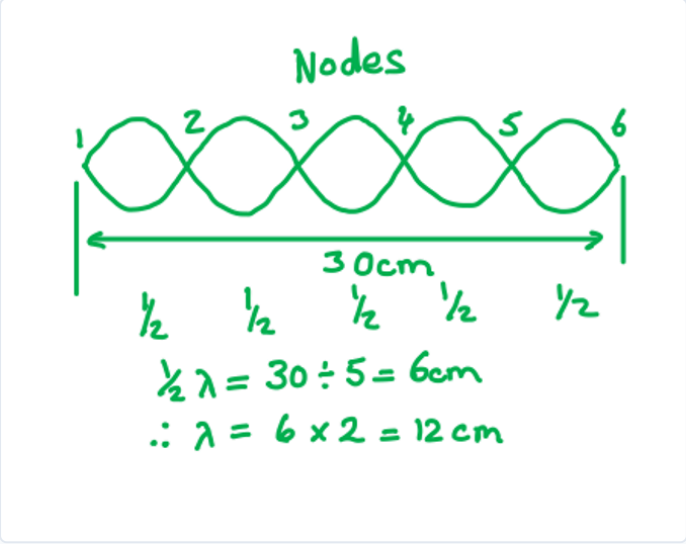
C 6.0 cm

D 5.0 cm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- There are five nodes in 30 cm and each has half a wavelength between each one
  - 30 cm divided by 5 = 6 cm
  - So, $\frac{\lambda}{2} = 6.0 \quad \text{cm}$
- Therefore, λ = 12.0 cm



Another method you can use to determine the wavelength in this question is to count up the number of wavelengths from the diagram (2.5) and then divide 30 cm by this number.

Question 18 Medium [1 mark]

A stationary sound wave produced in a musical instrument, this instrument has the pipes open at one end and closed at the other.

If the length of the pipe was 10 m what it is the lowest frequency of sound when the speed of sound is 320 m s^{-1}?

A 32 Hz

B 16 Hz

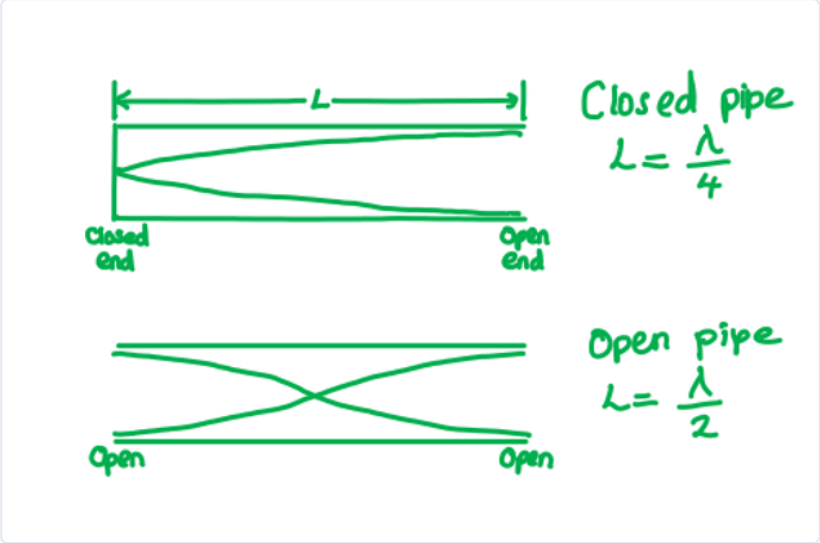
C 8 Hz

D 4 Hz

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The lowest or fundamental note produced will have only one node at the closed end and an antinode at the open end, creating ¼ of a wavelength

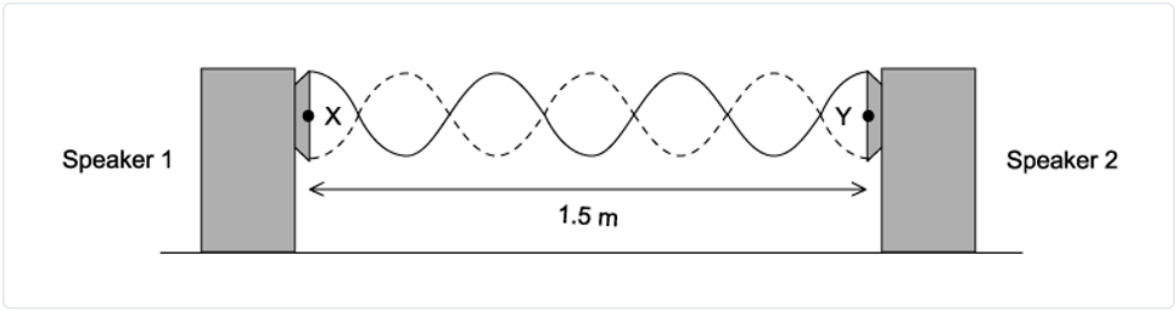


- A quarter of a wavelength is formed in a 10 m pipe
  - Therefore, one wavelength = 40 m
- The wave equation is given by $v = f \quad \lambda$

- $f = \frac{v}{\lambda} = \frac{320}{40} = 8 \quad \text{Hz}$

Question 19 Medium [1 mark]

Two loudspeakers emitting sound of the same frequency produces a stationary wave. This is shown in the diagram



The microphone is moved between X and Y for a distance of 1.5 m. There were six nodes and seven antinodes observed.

What is the wavelength of the sound?

A 0.21 m

B 0.25 m

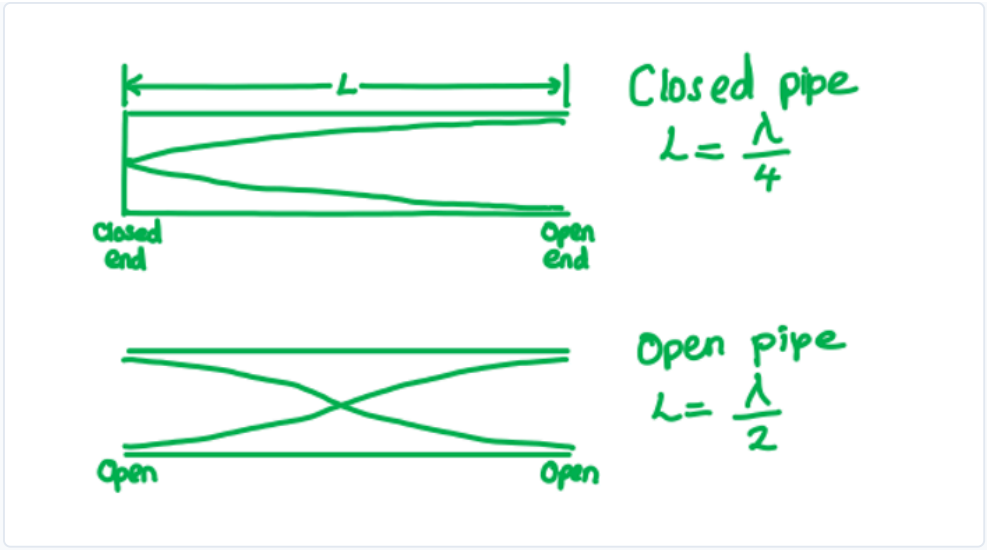
C 0.43 m

D 0.50 m

Mark Scheme & Explanation:

*The correct answer is **D** because:*

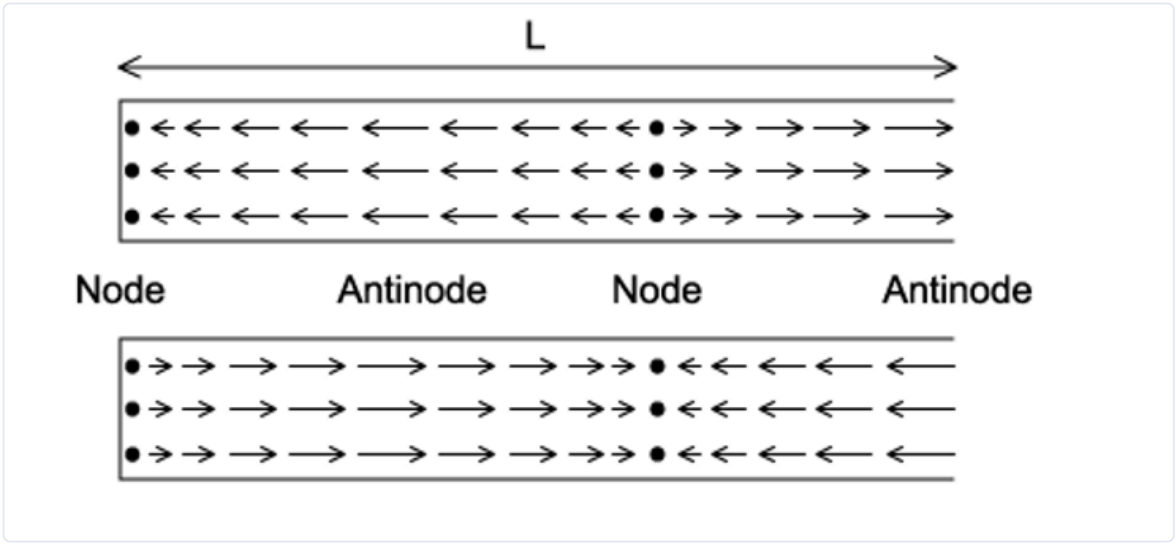
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude



- In this open-ended pipe, there are six nodes in 1.5 m each have half a wavelength between each one
    - 1.5 m divided by 6 = 0.25 m
    - So, $\frac{\lambda}{2} = 0.25 \text{ m}$
- Therefore, λ = 0.5 m

Question 20 Medium [1 mark]

A stationary wave was formed in a column. The diagram below shows the movement of the air particles.



The first diagram shows the displacement and the second is the displacement half a cycle later.

Which position will the pressure change by the most significant amount, and what is the length, L of the column in terms of wavelength, λ?

	Length L	Maximum pressure change
A	$\frac{3}{4}\lambda$	antinode
B	$\frac{3}{4}\lambda$	node
C	$\frac{3}{2}\lambda$	antinode
D	$\frac{3}{2}\lambda$	node

A

B

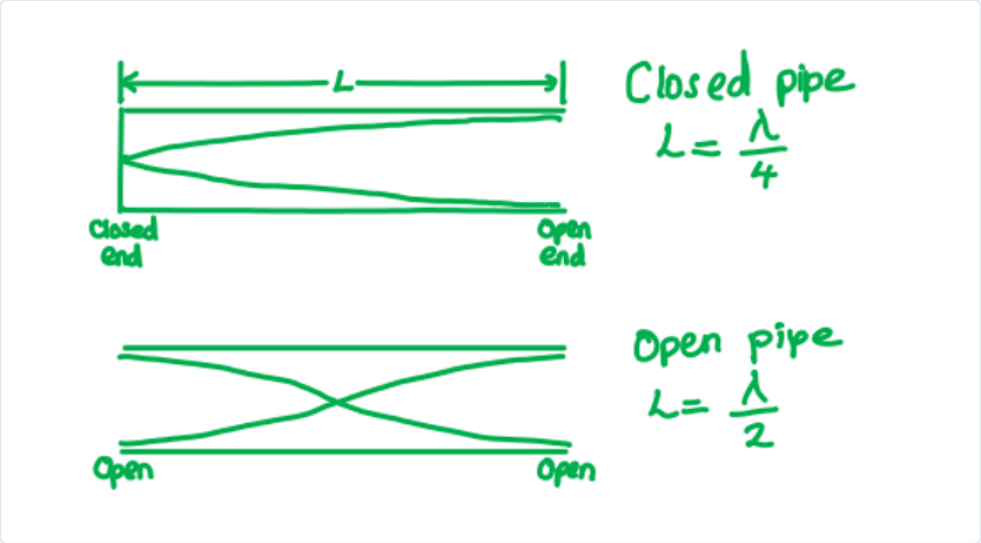
C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The pipe is closed at one end; there are two nodes present, therefore, the length of the tube is equal to $\frac{3}{4}\lambda$



- The maximum pressure change happens at the displacement node, where the particles are stationary, but density fluctuates the most

Question 21 Medium [1 mark]

A standing wave is formed in a closed pipe. The distance between the first node and the ninth node is 18.0 cm.

What is the wavelength of the wave?

A 4.5 cm

B 9.0 cm

C 12.0 cm

D 24.0 cm

Mark Scheme & Explanation:

The correct answer is A

- There are 9 nodes
- The distance between adjacent nodes is $\frac{\lambda}{2}$
- From node 1 to node 9, there are 8 node–node intervals

$$\frac{1}{2}\lambda = \frac{18.0}{8} = 2.25 \text{ cm}$$

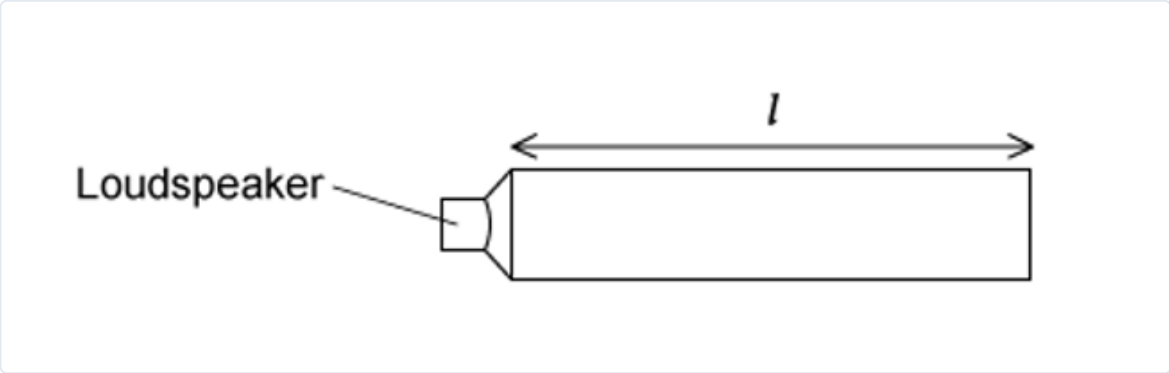
$$\frac{\lambda}{2} = 2.25 \text{ cm}$$

- Therefore

$$\lambda = 2.25 \times 2 = 4.50 \text{ cm}$$

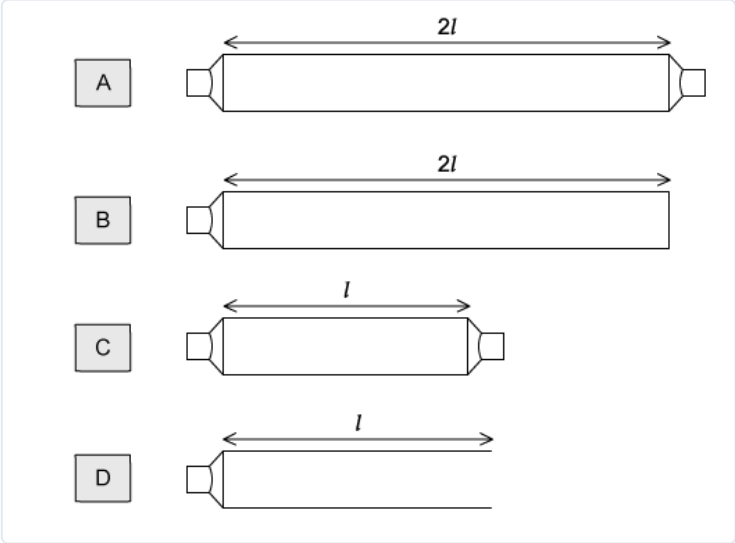
Question 22 Hard [1 mark]

A demonstration of a standing wave is set up using a loudspeaker. The loudspeaker is emitting sound with a frequency *f* and is placed at one end of the pipe with length *l*. The pipe is closed at the other end. This is shown in the diagram below.



Different pipes were set up with either one or two loudspeakers of frequency *f*. The pairs of loudspeakers were in phase with each other.

Which pipe would contain a standing wave ?



A

B

C

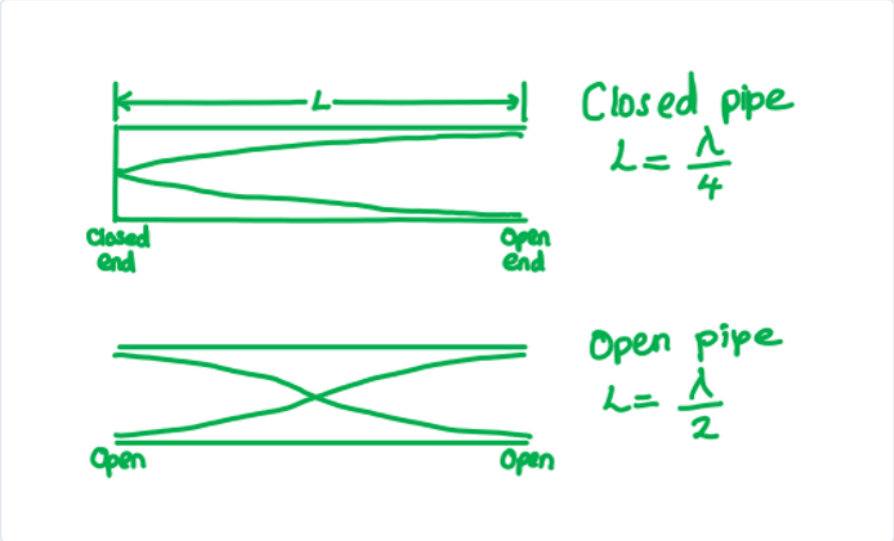
D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- For a stationary wave to be produced, there should be a node at the fixed end and an antinode at the loudspeaker

- For the demonstration in the question, the frequency is *f* and formed in the length of pipe *l*, for a fundamental mode the wave formed is a quarter of a wavelength
    - For a closed pipe :
    - $L = \frac{\lambda}{4}$ gives a wavelength of $\lambda = 4L$



- Option **A** shows a pipe with length $2L$ with two open ends
    - This would create half a wavelength, for this case, $\frac{\lambda}{2} = 2L$
    - Hence the wavelength is $\lambda = 4L$, which is the same as the demonstration in the question

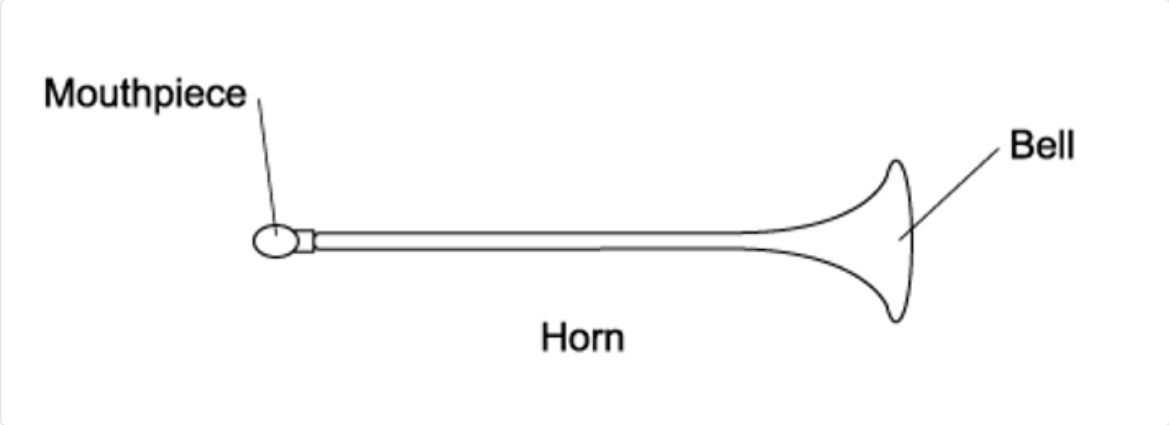
B is incorrect because a node will form at the closed end of the pipe and an antinode at the speaker, this is a quarter of a wave $\frac{\lambda}{4} = 2L$, therefore, $\lambda = 8L$

C is incorrect because there will be an antinode at both the speakers giving half a wavelength $\frac{\lambda}{2} = L$, therefore, $\lambda = 2L$

D is incorrect because there would be an antinode at the speaker and an antinode at the open end giving half a wavelength $\frac{\lambda}{2} = L$, therefore, $\lambda = 2L$

Question 23 Hard [1 mark]

When a wind instrument is used to create a note, a stationary wave is produced in the column. An example of this is the horn.



For the range of notes produced by the horn, a node is formed at the mouthpiece and an antinode is formed at the bell. The frequency of the lowest note is 75 Hz.

What would be the frequencies of the next two higher notes for this air column?

	First higher note / Hz	Second higher note / Hz
A	225	375
B	150	300
C	150	225
D	113	150

A

B

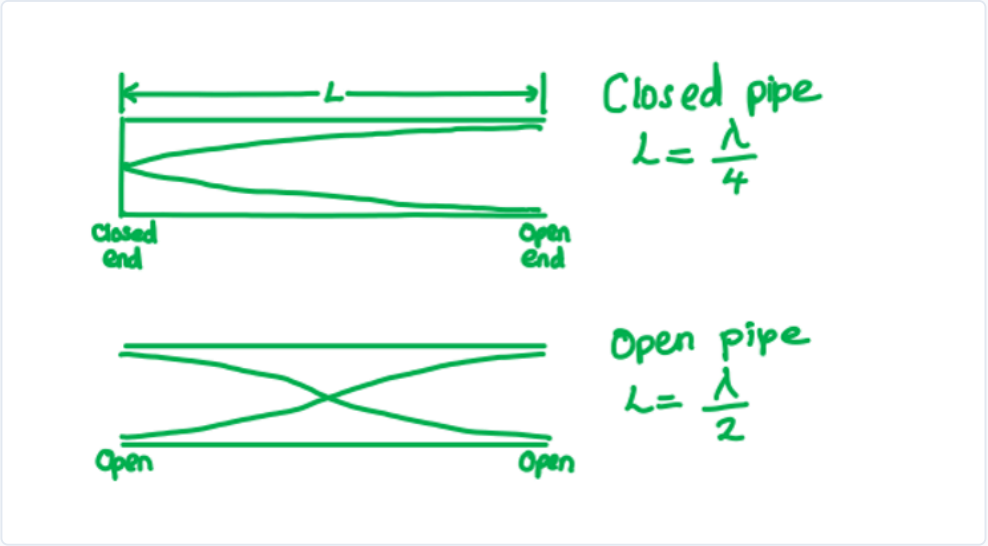
C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The question states that a node is formed at the mouthpiece and an antinode at the bell of the horn
- The lowest or fundamental note produced will have only one node at the mouthpiece and an antinode at the bell, creating a quarter of a wavelength

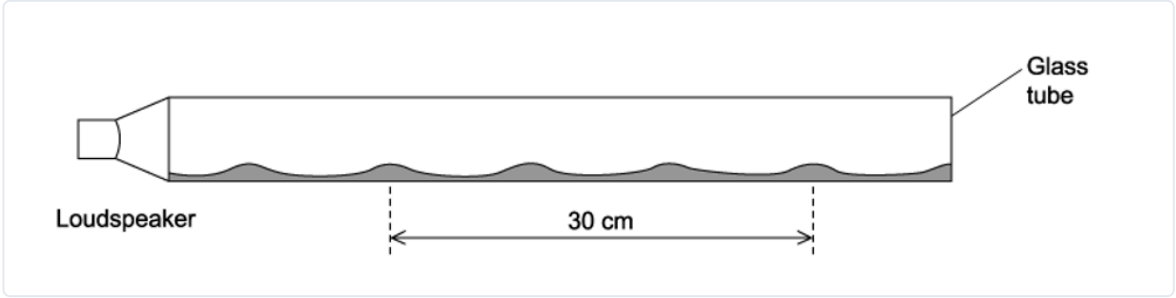


- The first higher note will have a node at the mouthpiece, then a node, followed by an antinode at the bell, giving $\frac{3}{4}$ of a wavelength
    - So, the frequency is 3 times that of the lowest note
    - Therefore, frequency $= 3 \times 75 = 225 \quad \text{Hz}$
- The second-highest note would be $\frac{5}{4}$ of a wavelength
    - So, the frequency is 5 times that of the lowest note
    - Therefore, frequency $= 5 \times 75 = 375 \quad \text{Hz}$

Question 24 Hard [1 mark]

A glass tube closed at one end, with a layer of fine powder laid inside as shown in the diagram, was set up. Sound was emitted from the loudspeaker and placed near the open end of the tube.

The frequency of the sound wave was changed, at one frequency a stationary wave was formed, and the powder formed 4 heaps. The distance between the four heaps of powder was 30 cm.



The speed of sound in the tube is 330 m s^{-1}

What is the frequency of the sound?

A 1650 Hz

B 2200 Hz

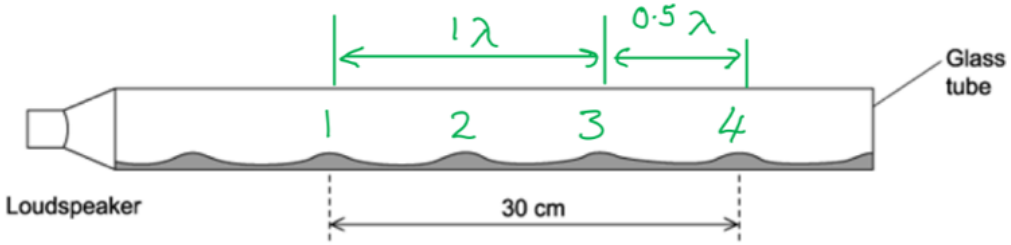
C 3300 Hz

D 6600 Hz

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The pile of dust will correspond to a node
- One wavelength is equal to three nodes, so, the distance formed by three successive heaps is equal to one wavelength

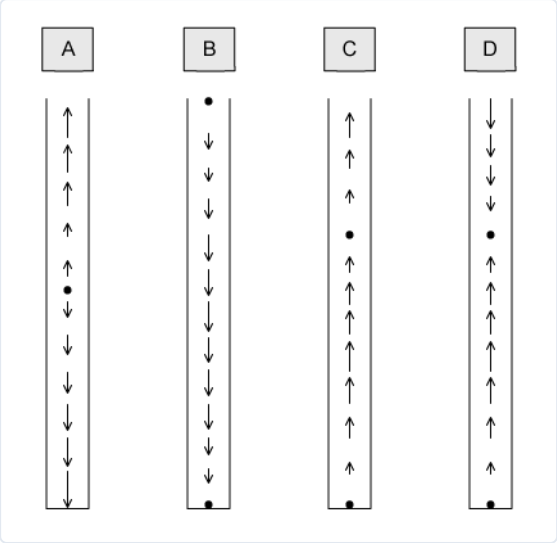


- So, the wavelength of the sound wave is :
  - $\frac{3}{2}\lambda = 0.3$
  - $\lambda = 0.3 \times \frac{2}{3} = 0.2 \quad \text{m}$
- The wave equation is given by $v = f\lambda$
  - Therefore, frequency, $f = \frac{v}{\lambda} = \frac{330}{0.2} = 1650 \quad \text{Hz}$

Question 25 Hard [1 mark]

A pipe is used to set up a stationary longitudinal wave. The length of each arrow in the diagrams below shows the motion of the air molecules. The arrowheads show the direction of the motion of the particle.

Which diagram shows a stationary wave with two nodes and two antinodes?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The waves are moving in opposite directions, and the standing wave results from alternately constructive and destructive interference
- A node is represented by a dot on the diagram, which shows zero displacement
- The arrows show the direction of travel of the wave
    - Hence diagram **D** shows two nodes and two antinodes

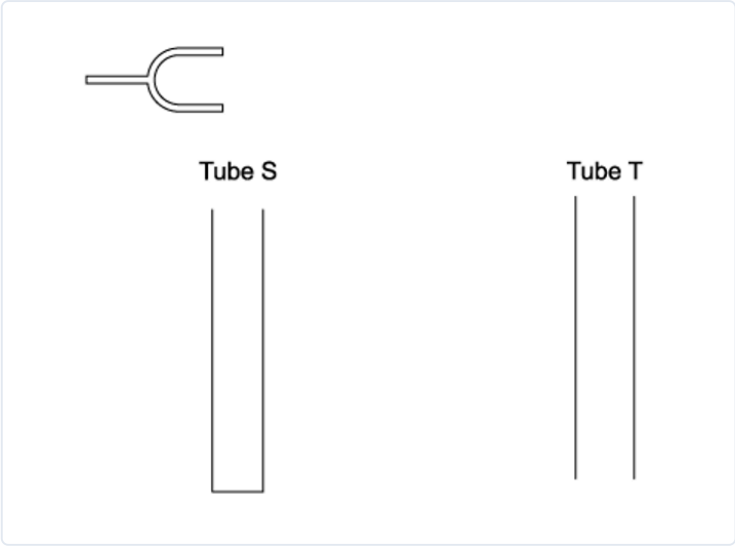
A is incorrect because the diagram only shows one node

B is incorrect because the diagram only shows one antinode

C is incorrect because a stationary wave is produced by the superposition of the wave travelling in the opposite direction. This diagram shows the waves all travelling in the same direction

Question 26 Hard [1 mark]

A student was investigating a tuning fork over two tubes. The tube were identical in length and size except tube S is closed at one end, and tube T is open at the lower end.



The tuning fork was placed above tube S and caused a resonance of the air at frequency *f*. No resonance is found at any lower frequency than *f* with tube S.

Which tuning fork will produce resonance when placed above tube T?

A a fork of frequency $2f$

B a fork of frequency $\frac{3f}{2}$

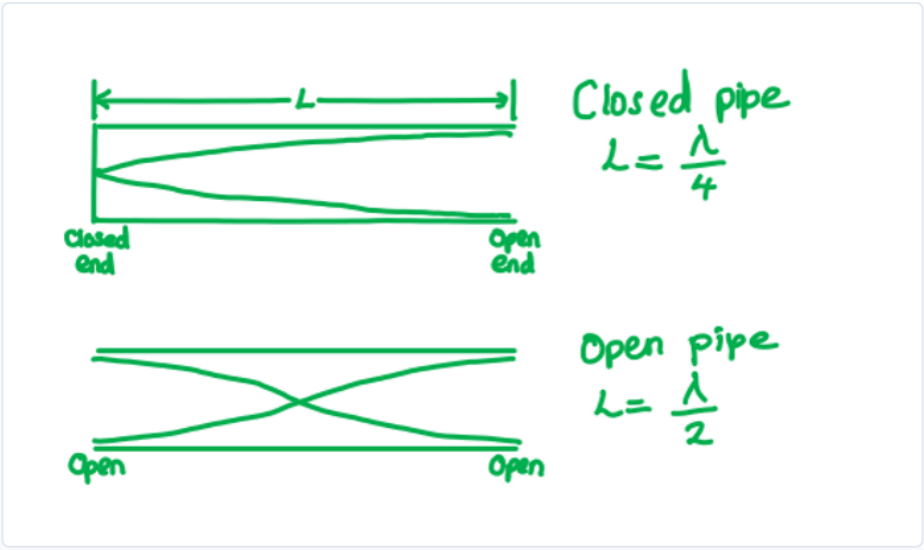
C a fork of frequency $\frac{2f}{3}$

D a fork of frequency $\frac{f}{2}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

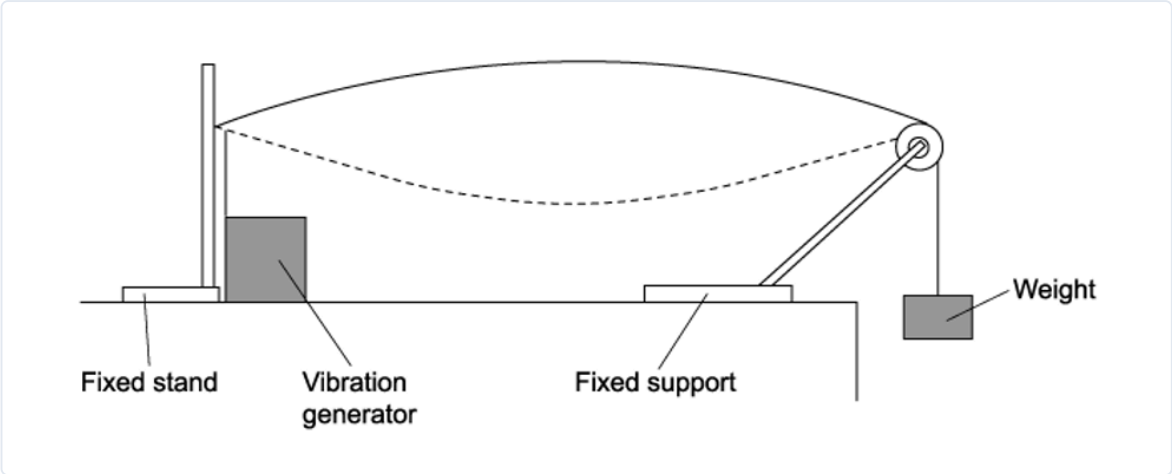
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- Tube S is a close-ended pipe, with a node at the closed-end, and an antinode at the open end
- Tube T is an open-ended pipe, with an antinode at both ends



- In a closed tube, the lowest or fundamental note produced will have only one node at the fixed end and an antinode at the open, creating a quarter of a wavelength
    - So $L = \frac{\lambda}{4}$, giving, $\lambda = 4L$
- In an open tube, the lowest or fundamental note produced will have an antinode at each end with a node in the middle, creating half of a wavelength
    - So $L = \frac{\lambda}{2}$, giving, $\lambda = 2L$
- For the closed tube S, the resonant frequencies are equal to:
    - $f_c = \frac{nv}{4L}$
    - Where v = speed of the wave, L = length of the tube and n = 1,3,5...
- For the open tube T, the resonant frequencies are equal to:
    - $f_o = \frac{nv}{2L}$
    - Where n = 1,2,3...
- So, the resonant frequency for the open tube T (when n = 1):
    - $f_o = 2f_c$

Question 27 Hard [1 mark]

A steel wire was set up clamped at one end and placed under tension by a mass hung over a pully; this is shown in the diagram.



A vibration generator is attached to the wire nearest to the clamped end. A stationary wave with one loop is produced. The frequency of the generator is *f*.

Which frequency should be used to produce two loops?

A $2 \quad f$

B $4 \quad f$

C $\frac{f}{2}$

D $\frac{f}{4}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- In the diagram 1 loop is formed (1st harmonic) and this is half a wave so on the length of the wire, *L*
  - $L = \frac{\lambda_1}{2}$ so, wavelength $\lambda_1 = 2L$
  - Where λ is the wavelength of the wave
- The wave equation is given by $v = f\lambda$
  - So, frequency, $f_1 = \frac{v}{\lambda_1} = \frac{v}{2L}$

- For 2 loops to be formed (2nd harmonic) on the same length of wire:
    - $\lambda_2 = L$, as a complete wave will be formed
- Therefore, the new frequency is equal to :
    - $f_2 = \frac{v}{\lambda_2} = \frac{v}{L} = 2f_1 = 2f$

Question 28 Hard [1 mark]

An investigation to sound waves reflected by a wall was carried out. The frequency *f* of the wave was adjusted until a stationary wave is formed with an antinode with a distance *x* closest to the wall.

Which expression gives *f* in terms of *x* and the speed of sound *c* ?

A $f = \frac{c}{2x}$

B $f = \frac{c}{4x}$

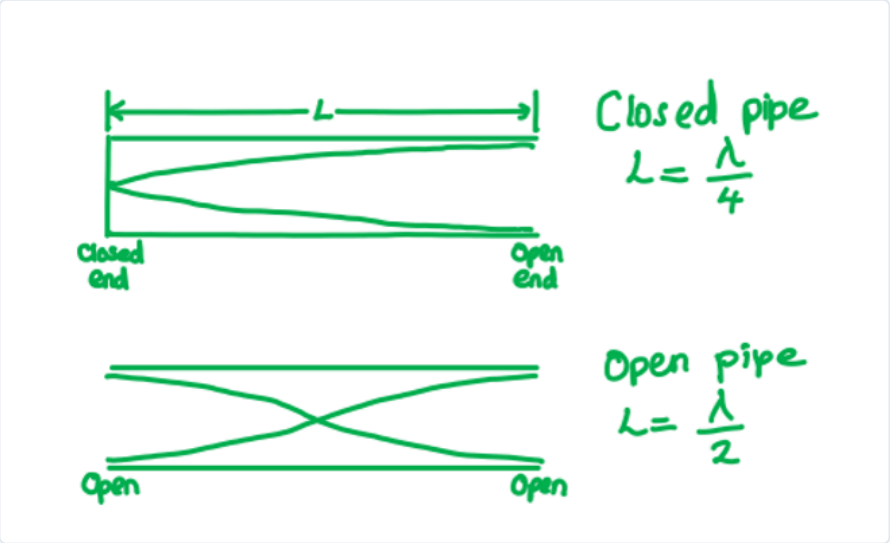
C $f = \frac{2c}{x}$

D $f = \frac{4c}{x}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A node would be present at the wall, the same as in a closed tube

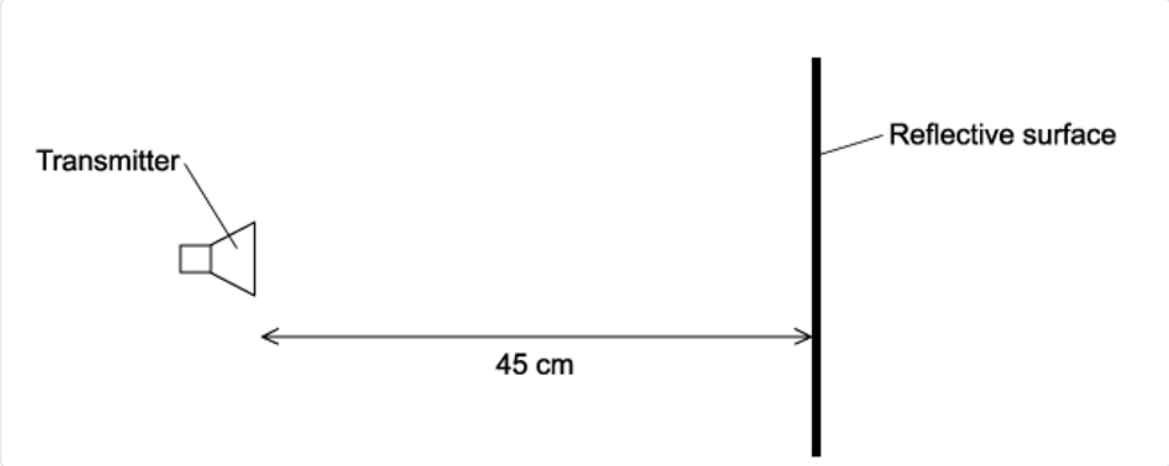


- The distance between a node and antinode is equal to $\frac{\lambda}{4}$
  - So, $\frac{\lambda}{4} = x$, so, wavelength $\lambda = 4x$
- The wave equation is equal to $c = f\lambda$

- Therefore, frequency : $f = \frac{c}{\lambda} = \frac{c}{4x}$

Question 29 Hard [1 mark]

A student was investigating the reflection of electromagnetic waves. They set up a transmitter 45 cm from a reflective surface, as shown in the diagram.



The waves had a frequency of 1.00 GHz. A stationary wave was produced with a node at the transmitter and a node at the surface.

How many antinodes are there in the space between the surface and the transmitter?

A 4

B 3

C 2

D 1

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The wave equation is equal to $c = \lambda f$
- Where:
  - Speed of electromagnetic waves : c = 3.0 × 10^8 m s^{-1}
  - Frequency : f = 1 GHz = 1 × 10^9 Hz
  - Wavelength : $\lambda = \frac{v}{f} = \frac{3.0 \times 10^8}{1 \times 10^9} = 0.3$ m

- The distance between the transmitter and the reflective surface = 45 cm
    - So, the number of periods $= \frac{0.45}{0.30} = 1.5$
- One antinode is half a period, so, in 1.5 periods there are 3 antinodes

Question 30 Hard [1 mark]

Stationary sound waves can be produced in an organ pipe of length L when both ends are open, and this creates a note from an organ.

The speed of sound in the air column is v.

What is the lowest (fundamental) frequency of the note produced in the pipe?

A $\frac{v}{4\mathrm{L}}$

B $\frac{v}{2\mathrm{L}}$

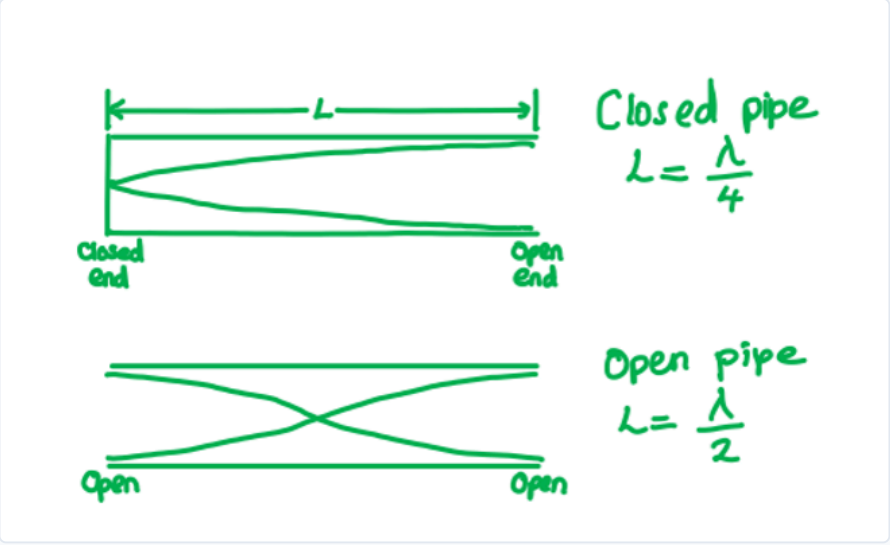
C $\frac{v}{\mathrm{L}}$

D $\frac{2v}{4\mathrm{L}}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The organ pipe is open at both ends, in order for it to have the lowest (fundamental) frequency it will have two antinodes, with a single node in the middle

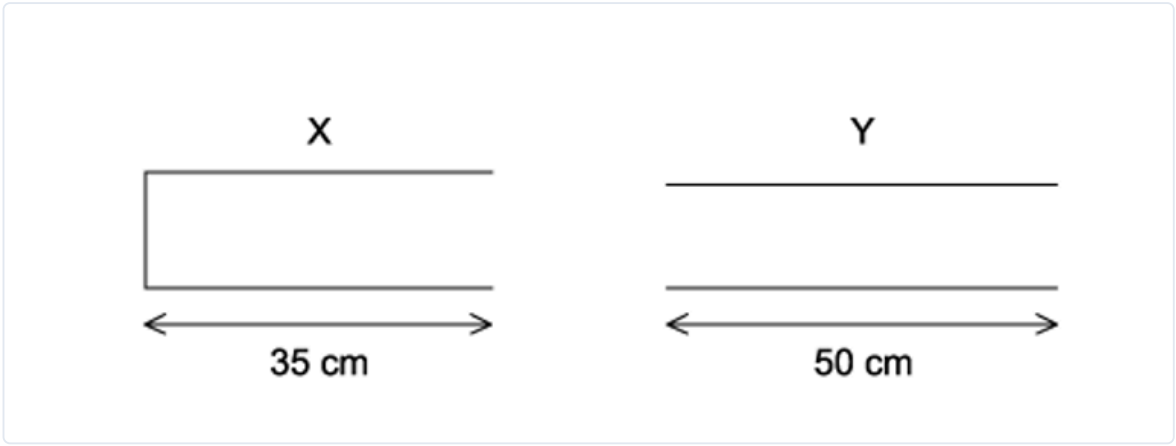


- The wavelength can be measured by measuring the length between 3 consecutive nodes, the distance between the 1st and 2nd nodes give half the wavelength
    - So, in the organ pipe $\mathrm{L} = \frac{\lambda}{2}$

- Speed of a wave, $v = \lambda f$
  - Therefore, the fundamental frequency is $f_f = \frac{v}{\lambda} = \frac{v}{2L}$

Question 31 Hard [1 mark]

A student was investigating the creation of stationary waves. They set up travelling waves with a wavelength of 20 cm in a closed pipe X and an open pipe Y. The lengths of the pipes are shown in the diagram.



In which pipe did the student observe stationary waves?

A X only

B X and Y

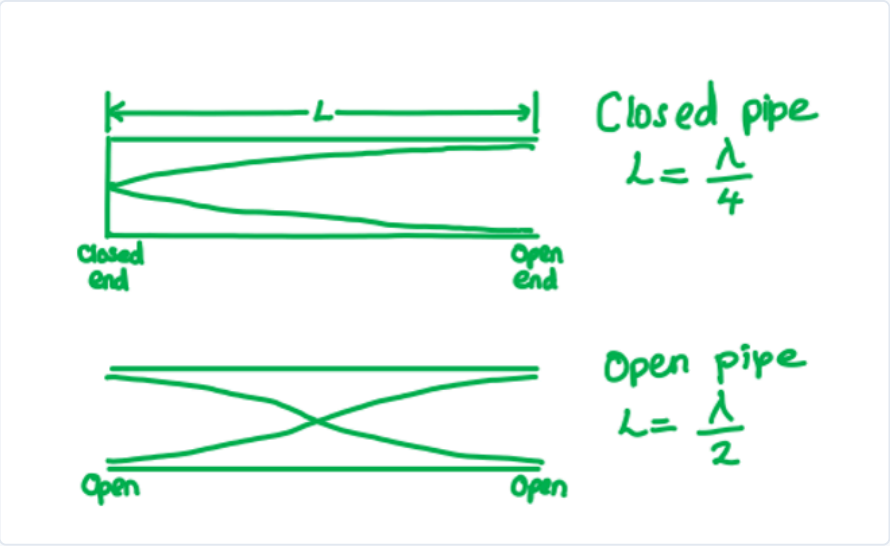
C Y only

D neither X or Y

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude



- In an open-ended pipe there will be half a wavelength, so for pipe Y:
  - $L = \frac{x}{2} \times \lambda$
  - $50 = \frac{x}{2} \times 20$
  - $x = 5$
- There is an odd whole number produced
  - Therefore, a standing wave will be created
- In a close-ended pipe there will be a quarter of a wavelength, so for pipe X:
  - $L = \frac{x}{4} \times \lambda$
  - $35 = \frac{x}{4} \times 20$
  - $x = 7$
- There is an odd whole number produced
  - Therefore, a standing wave will be created
- So, both X and Y will produce a standing wave